

SIADS 696 Milestone II Project Report

T-cell Receptor and Antigen Binding Prediction

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1: Introduction

T-cell receptor (TCR) therapy is a promising strategy for cancer treatment utilizing the body's immune system to remove cancer cells^{1,2}. T-cells are responsible for identifying and eliminating infected or cancerous cells. The T-cells identify their targets through the interaction of the T-cell receptors and a specific antigen³. This antigen is taken from protein fragments in the cell then presented on the surface for evaluation by the T-cell receptors⁴. If there is a match, the cell will be targeted for elimination by the immune system. Modern computational methods have been developed to virtually screen for novel cancer epitopes and TCR binding sequences that can identify them^{5,6}. However, the available options for targeted cancer treatments remain extremely limited. Our project provides a new method for predicting binding between the TCR binding domain and the target antigen epitope for the development of novel targets for new therapies.

Drug discovery and development is a costly and time-consuming process. The methods developed in this project will provide resources to reduce the time and overhead by identifying high priority targets. This process could effectively halve the lab work required because our method uses identified antigens as the fixed target so TCR binding domains can be developed and assessed in the virtual environment.

Cancer is something that has affected virtually everyone's life in some way. TCR and CAR-T therapies represent the current greatest advancement in the treatment of cancer. Modern machine learning techniques utilized here provide a means to accelerate and improve the development of these life-changing technologies. TCR binding domain optimization can be reframed as a sequence-pairing problem between the TCR peptide sequence and the target epitope sequence. Since both can be represented as one-dimensional sequences and both are represented as sequences of discrete tokens (amino acids or words), this task is conceptually similar to challenges in Natural Language Processing (NLP) and allows for the application of Natural Language Processing (NLP) and Large Language model techniques to analyze protein sequences

Therefore, aiming to use ML to predict binding between TCR peptide sequence and the target epitope sequence, collected protein sequence dataset from the VDJdb (<https://vdjdb.cdr3.net/>), and leveraged NLP techniques and unsupervised learning to extract useful features from them, and used those features in subsequent supervised learning tasks

Specifically, for unsupervised learning tasks, we compared two feature engineering methods for data preprocessing and two dimensionality reduction techniques for feature extraction. We then applied a clustering algorithm to identify distinct groups within the data. Our results demonstrate that a protein language model provides a superior representation of protein sequences compared to one-hot encoding. Furthermore, we found UMAP to be more suitable for dimensionality reduction of this data than PCA. Ultimately, we developed 97 new features (88 from PCA, 8 from UMAP, and 1 from K-means clustering) for use in downstream supervised learning tasks.

For supervised learning phase of this project aimed to develop models capable of predicting T-cell receptor (TCR) and epitope binding interactions using a labeled dataset derived from the VDJdb database. Our supervised learning integrates the enriched representations—including protein language model embeddings (ESM1v), PCA and UMAP projections, and cluster features—to provide a comprehensive and high-dimensional view of sequence and structural information. The successful goal of this analysis was to assess model performance, evaluate the impact of feature reduction, and identify which engineered features most influence predictions.

In summary, our project successfully optimized an unsupervised and supervised pipeline for protein sequence feature extraction and prediction. Our results demonstrate the advantage of using a protein language model over traditional one-hot encoding and establish the combination of non-linear dimensionality reduction and clustering as an effective method for analyzing protein sequences.

2: Related Works

Machine learning has proven to be a powerful and efficient tool for analysis and prediction of protein interactions. Chemical and structural based approaches, like the one used by Dhusia et al in 'A structural-based machine learning method to classify binding affinities between TCR and peptide-MHC complexes' demonstrate the viability of using TCR sequences to predict their binding⁷. While this method is relatively effective, our project focuses on an additional evaluation method by evaluating the amino acid sequences directly with NLP.

The primary inspiration for our project was the paper "Predicting TCR sequences for unseen antigen epitopes using structural and sequence features" by Ji et al (2024)⁸, which utilized transformer and convolutional neural networks to identify sequence features for predictions. This method utilized a generative model that pitted a pretrained epitope encoder system against a transformer decoder for predicting CDR3-beta binding sequences. Our project reverses the focus by looking at the antigen, which is a fixed value, rather than TCR sequences which are the target for therapy development.

The work described in "Comparative Analysis of TCR and CAR Signaling Informs CAR Designs With Superior Antigen Sensitivity and In Vivo Function"⁹ demonstrated the type of result we would hope to see from our model. They utilized a combination of CAR-T and TCR receptor constructs to identify potential targets for improvement. The method developed in this report could be utilized to evaluate the resulting constructions to determine the optimal candidates for drug development.

3: Data Sources

The VDJdb (<https://vdjdb.cdr3.net/>) is a publicly available database of T-cell receptor sequences with their known antigen targets¹⁰ which served as the primary dataset for our analysis. The entire dataset was downloaded as a .tsv file of ~140mb containing 20K TCR/antigen paired sequences and their supporting features.

The primary variables from this dataset were the TCR binding domain amino acid sequence ('CDR3'), the antigen epitope amino acid sequence ('antigen.epitope'), and the binding score. These amino acids sequences represent chain molecules with specific properties functioning similarly to how words form a sentence. The score parameter ranged from 0-3. Scores of 1, 2, or 3 represent increases in the confidence of the binding between the TCR sequence and the antigen ranging from moderate to exceedingly sure. Scores of 0 are given when there is not enough information to rate the confidence in the binding.

4: Feature engineering

The feature engineering pipeline followed the following steps:

1. **Data Extraction and Cleaning:** The three primary variables were extracted and filtered for duplicates, reducing the Dataframe from 180321 rows x 22 columns to 113729 x 3.
2. **Create the Negative Population:** All reported pairings are suspected to have some level of binding, meaning there were no true negative samples to predict the lack of binding. The dataset was partitioned with samples the (insert number here) with a score above 0 serving as the positive samples. An equal number of randomly selected samples of the remaining population were taken to serve as the negative population. The TCR sequences were randomly assigned to antigen sequences, evaluated to ensure novel binding pairs, then assigned to the negative population creating 18,864 protein sequence pairs.
3. **PyProtein Feature Extraction:** The python module PyProtein contained in PyBioMed was used to extract an additional 147 physical/chemical characteristics for each sequence.
4. **One-Hot Encoding:** method creates a sparse, high-dimensional binary vector where a single '1' indicates the presence of a specific category (an amino acid). It treats all categories as independent. For our data, this involved creating a vocabulary of the 20 unique amino acids and representing each protein sequence as a vector of amino acid frequencies. This resulted in a 20-dimensional vector for each sequence, analogous to a "Bag of Words" representation in NLP.

5. **Evolutionary Scale Modeling (ESM-1v):** ESM is a transformer-based protein language model developed by Meta AI. This model learns rich, dense vector representations of protein sequences from large-scale protein sequence datasets, capturing biological features directly from the raw sequences without requiring explicit structural information. In contrast to one-hot encoding, these embeddings are dense and capture semantic and contextual similarities, positioning similar sequences closer in the vector space. This method generated a 1,280-dimensional representation for each sequence.
6. **Sequence Embedding Extraction:** Preprocessing was performed by applying one-hot encoding to the amino acid sequences for the TCR and antigen. Each sequence was also inferred using ESM model to get protein-level embedding.

<u>Column Name</u>	<u>Description</u>	<u>Data Type</u>	<u>Data Source</u>
cdr3	Amino acid sequence for TCR binding domain	text	VDJdb
antigen.epitope	Amino acid sequence for target antigen	text	VDJdb
vdjdb.score	The rating for the quality of the binding prediction	numerical	VDJdb
Binding	Predicted binding of the cdr3 and antigen pairing	numerical	Engineered from VDJdb sequences
embedding	Column containing the one-hot encoding features	numerical	Engineered from VDJdb sequences
_emb_dim	Columns containing the ESM dimensions	numerical	Engineered from VDJdb sequences
features	Columns containing the unsupervised learning derived features	numerical	Engineered from '_embedding_' or '_emb_dim'

Table 1: List of data columns used throughout the model development.

5: Unsupervised Learning

Methods & Justification

Our unsupervised learning workflow was designed to discover the intrinsic grouping structure within the protein sequence data. The pipeline consisted of three integrated stages: feature engineering (discussed above), dimensionality reduction, and clustering

1. Feature Engineering

The comparison and justification for using PLM over one-hot encoding were detailed previously. In short, we generated protein sequence representations using two methods: one-hot encoding and ESM-1v, a protein language model. One-hot encoding creates a simple, 20-dimensional vector representing amino acid frequencies, treating each amino acid independently. ESM-1v embeddings are dense, 1,280-dimensional vectors learned from large datasets that capture biological and contextual similarities in sequences.

2. Dimensionality Reduction

We employed two distinct dimensionality reduction techniques, Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP), to visualize the data structure in two dimensions and to create refined feature sets for clustering. PCA is a linear technique that projects data onto orthogonal axes (principal components) to capture the maximum global variance. It is computationally efficient and highly interpretable, making it ideal for identifying the main directions of variance in the data. UMAP is a non-linear manifold learning technique designed to preserve both the local and global topological structure of the data, assuming it lies on a lower-dimensional manifold. UMAP is particularly powerful for visualizing complex cluster structures that linear methods like PCA might not capture, and widely used in biomedical research. By comparing PCA and UMAP, we directly contrast linear and non-linear dimensionality reduction approaches for protein sequence data, allowing us to determine which method better reveals the underlying clustering structure.

3. Clustering

We applied the K-Means algorithm to the reduced feature spaces generated by both PCA and UMAP to identify distinct customer segments. K-Means is a centroid-based, scalable clustering algorithm that partitions data by minimizing within-cluster variance, assigning each point to the nearest cluster center. K-Means was selected for its interpretability, computational efficiency, and effectiveness in identifying spherical, well-separated clusters, which aligns with our goal of discovering distinct groupings.

Results

1. Data Preprocessing

We first transformed the protein sequence data into numerical representations to facilitate analysis. For the 18,864 protein sequence pairs in our raw dataset, we generated two distinct datasets. The first used a bag-of-words approach, creating a 40-dimensional vector for each pair based on amino acid frequency. The second utilized the ESM-1v protein language model to generate a dense, 2,560-dimensional representation for each pair. These are referred to as the onehot and esm datasets, respectively, throughout the following experiments.

2. Dimensionality Reduction

Then, we performed hyperparameter tuning for both PCA and UMAP for dimensionality reduction. For PCA, we selected the number of components by analyzing the cumulative explained variance ratio, choosing a value that retained >95% of the variance (Figure 1). This resulted in 33 components for the onehot dataset and 88 for the esm dataset. For 2D visualization, both were projected into two dimensions, where the esm data exhibited a clearer clustering structure (Figure 2).

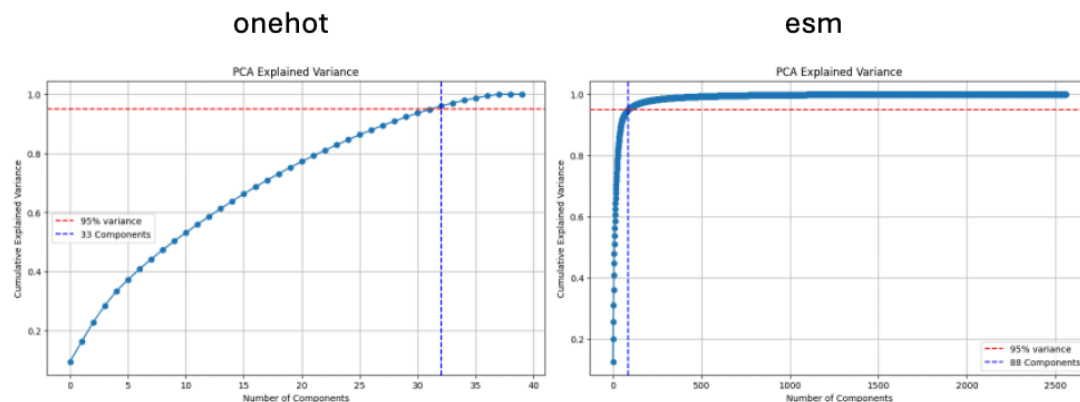


Figure 1. Selection of PCA Components Based on Cumulative Explained Variance Ratio

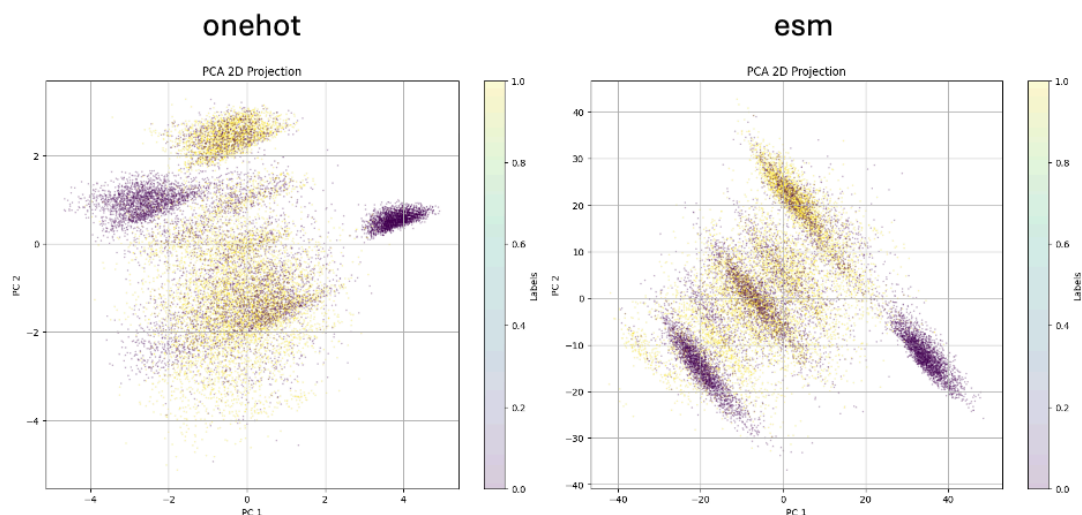


Figure 2. Two-Dimensional PCA Projections of One-Hot and ESM Datasets

For UMAP hyperparameter tuning, which cannot be explained by variance, we used Silhouette Score—a metric that measures how well-separated the resulting clusters are—to optimize the number of components (Figure 3). The onehot dataset achieved the highest score with 4 components, while the esm dataset required 8. A subsequent grid search for `n_neighbors` and `min_dist` found optimal parameters of (15, 0.1) for onehot (Silhouette Score: 0.1155) and (30, 0.1) for esm (Silhouette Score: 0.1390) (Figure 4). This 2D UMAP projections, colored by known labels, revealed that the esm representation could separate positive and negative data points into some distinct clusters, a structure not achieved by the onehot data(Figure 4).

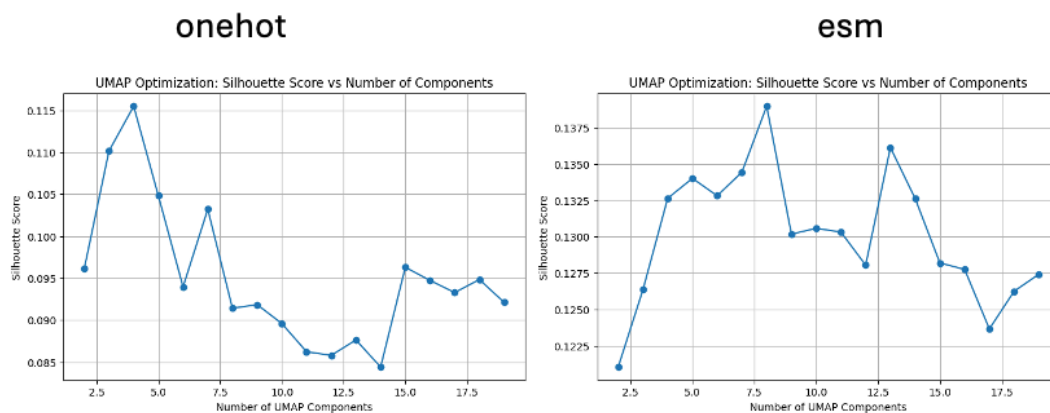


Figure 3. UMAP Hyperparameter Optimization via Silhouette Score

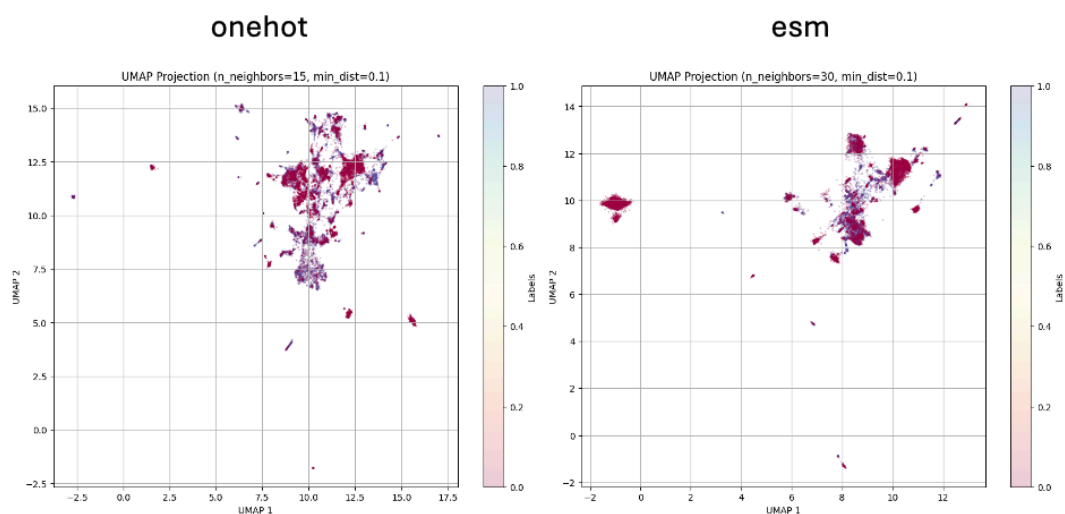


Figure 4. UMAP Parameter Grid Search for $n_neighbors$ and min_dist and Two-Dimensional UMAP Projections Colored by Known Labels

3. Clustering and Final Pipeline

Given UMAP's superior performance in revealing cluster structure, we used the UMAP-reduced features for clustering with K-Means. We determined the optimal number of clusters k using the Elbow Method and Silhouette Analysis, finding $k=22$ for the onehot dataset and $k=28$ for the esm dataset(Figure 5).

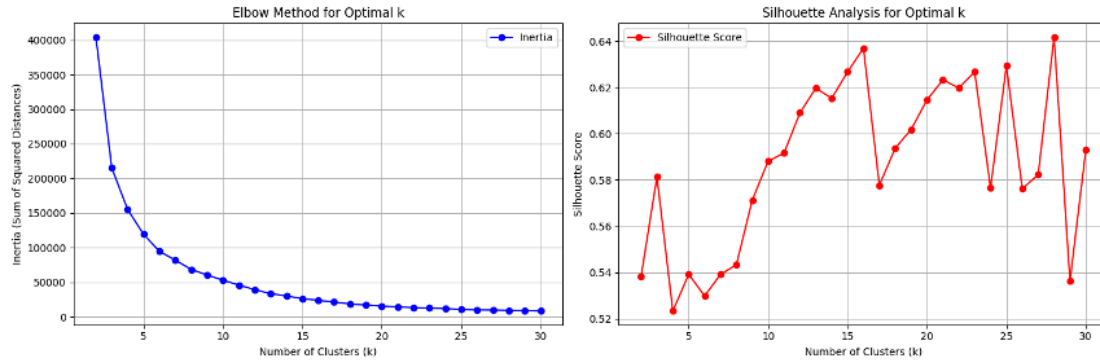


Figure 5. Determination of Optimal Cluster Number (k) by Elbow and Silhouette Methods for UMAP features

The final visualizations confirmed that the combination of the esm dataset, UMAP for dimensionality reduction, and K-Means clustering provided the most coherent and well-separated cluster visualization than onehot dataset and for dimensionality reduction. Consequently, this combination was selected as the final setting for unsupervised feature extraction(Figure 6).

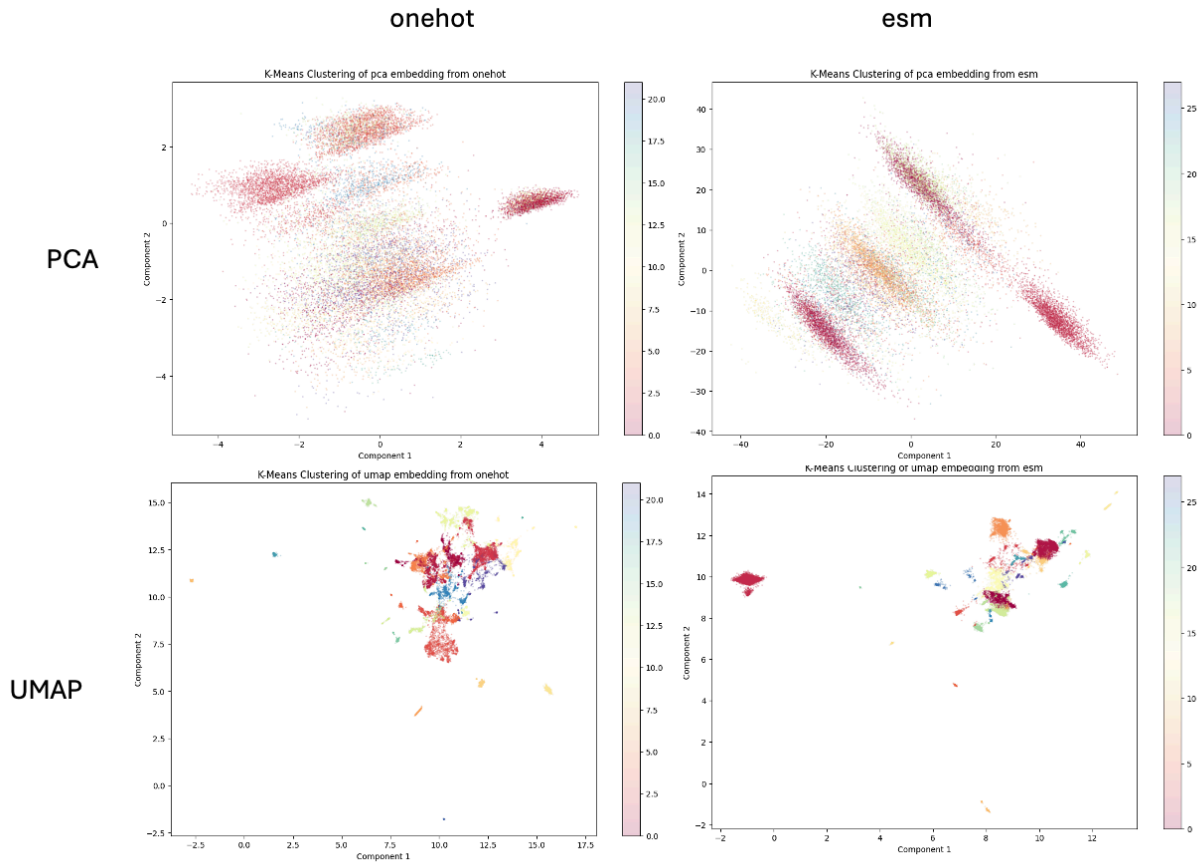


Figure 6. Comparison of Final Cluster Visualizations Using UMAP-Reduced Features

4. Sensitivity Analysis

We evaluated the sensitivity of our method to the key hyperparameters of UMAP ($n_neighbors$) and K-Means ($n_clusters$) by visualization of the clustering using different parameter (Figure 7). As show in the figure, the clustering stability was moderately sensitive to UMAP's $n_neighbors$. Low and medium values (15, 30) maintained a robust Silhouette Score of ~ 0.62 , while a high value (45) caused over-separation and a lower score (0.58). On the other hand, the model was highly sensitive to the $n_clusters$ parameter in K-Means. Deviating from the optimal $k=22$ —for example, using $k=12$ or $k=32$ —led to either the merging of distinct groups or over-segmentation, significantly reducing cluster quality and Silhouette Score (to 0.49), confirming that our identified optimum was robust.



Figure 7. Sensitivity Analysis of UMAP $n_neighbors$ and K-Means $n_clusters$ Parameters

In summary, we successfully optimized an unsupervised pipeline for protein sequence feature extraction. Our results demonstrate the advantage of using a protein language model over traditional one-hot encoding and establish the combination of non-linear dimensionality reduction and clustering as an effective method for analyzing protein sequences.

Discussion

Our unsupervised learning results show that embeddings from the protein language model perform better than one-hot representations. As the parameters indicate, the ESM embeddings contain more information. For example, the optimized parameter such as the number of components in UMAP is larger for ESM than for one-hot, and the ESM embeddings also require a larger number of clusters in k-means.

One of the major challenges is that ESM embeddings have much higher feature dimensions than one-hot representations, meaning a more complex model is needed for dimensionality reduction. We observed that the ESM UMAP model is significantly larger and slower than the one-hot version, so further optimization is required.

With extended time, we could first filter high-dimensional features from the protein language model to obtain a lower-dimensional representation and then perform unsupervised learning on the reduced features. This would make the model much faster. We will also explore other clustering methods and compare them with k-means to optimize the clustering performance.

6: Supervised Learning

Methods and Justification

Three model families were chosen to represent increasing levels of complexity and flexibility:

1. Logistic Regression (LR): A linear model providing interpretability and a baseline benchmark.
2. Random Forest (RF): A bagging ensemble capable of modeling nonlinear dependencies.
3. XGBoost (XGB): A gradient boosting algorithm optimized for accuracy and interpretability.

All models were trained on a standardized dataset using the top 200 features derived from earlier feature ranking and importance analysis. Each model was trained with 5-fold stratified cross-validation, optimizing for ROC-AUC during hyperparameter tuning via GridSearchCV. Model evaluation metrics included ROC-AUC, F1-score, Accuracy, Precision, and Recall. Both baseline and tuned versions of each model were compared to assess generalization and the effectiveness of hyperparameter optimization.

Results

Model Performance

All models achieved strong predictive results, with XGBoost and Random Forest outperforming Logistic Regression. Ensemble models demonstrated superior generalization to complex, nonlinear relationships in the high-dimensional embedding space (Figure 7).

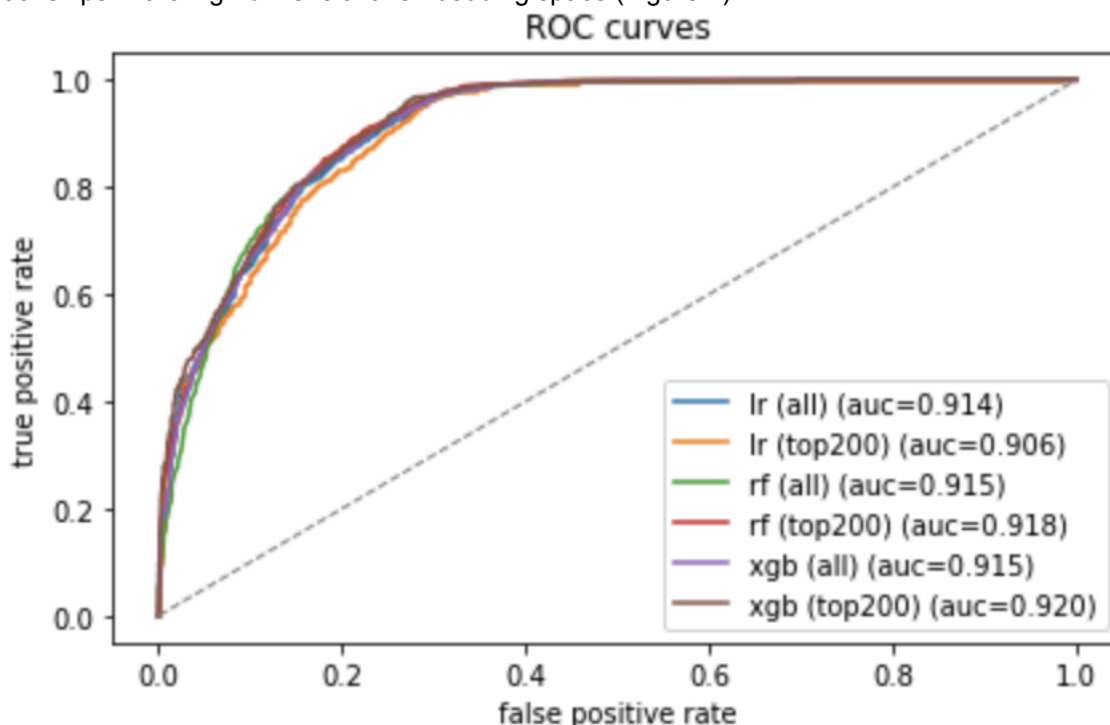


Figure 7. ROC Curves - Receiver Operating Characteristic (ROC) curves for all models comparing the full feature set (all) versus the reduced feature set (top 200). XGBoost achieved the highest overall AUC (0.920) with minimal overfitting.

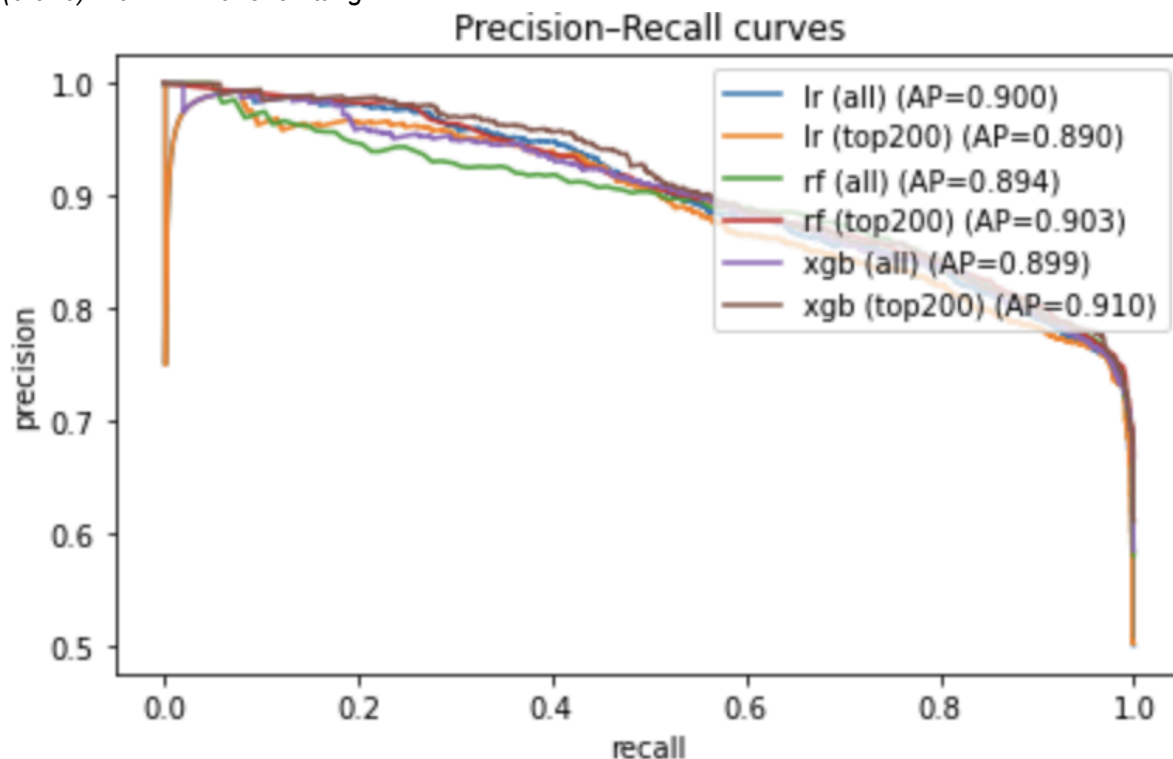


Figure 8. Precision-Recall Curves - Precision-Recall curves for Logistic Regression, Random Forest, and XGBoost models. Ensemble methods demonstrate higher average precision (AP $\approx$ 0.91), indicating better discrimination under class imbalance.

Table 1. 5-Fold Cross-Validation Results (Mean $\pm$ Std)

5-fold cross-validation (mean $\pm$ std)

	roc_auc	f1	accuracy	precision	recall
model					
logistic (top200)	0.915 $\pm$ 0.002	0.851 $\pm$ 0.003	0.837 $\pm$ 0.003	0.782 $\pm$ 0.006	0.932 $\pm$ 0.009
random forest (top200)	0.918 $\pm$ 0.001	0.849 $\pm$ 0.005	0.839 $\pm$ 0.005	0.802 $\pm$ 0.007	0.901 $\pm$ 0.012
xgboost (top200)	0.921 $\pm$ 0.002	0.858 $\pm$ 0.004	0.845 $\pm$ 0.006	0.794 $\pm$ 0.008	0.932 $\pm$ 0.004

Feature Importance

Feature importance was extracted from each model to determine the key variables contributing to predictive performance. Across models, protein language model derived epitope embedding dimensions and UMAP/PCA-derived dimensionality reduction features consistently appeared among the top-ranked features, demonstrating that sequence-derived embeddings effectively capture biologically relevant binding signals.

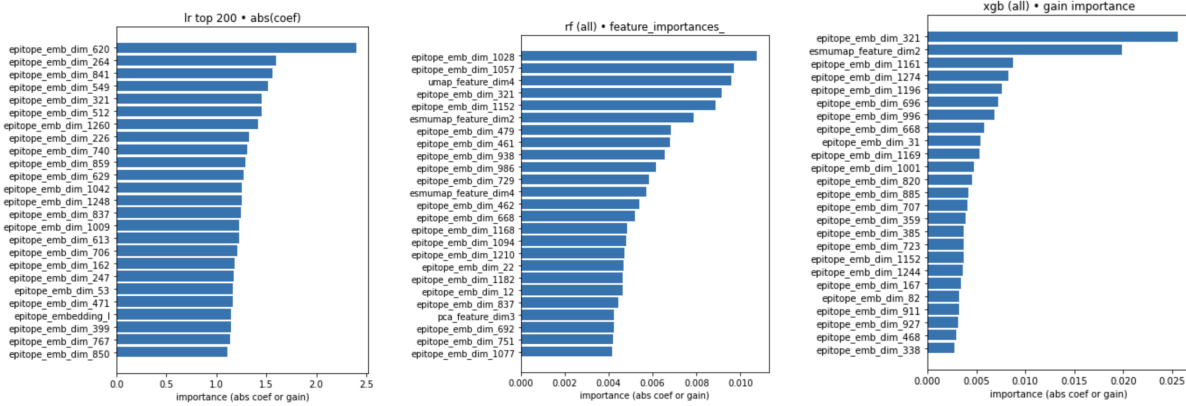


Figure 9: **A** Logistic Regression Feature Importance - Top 25 Logistic Regression coefficients (absolute value). Epitope embedding dimensions dominate, indicating linear separability within the embedding space. **B**. Random Forest Feature Importance - Random Forest feature importances showing top ESM and UMAP components. These nonlinear features capture broader structural relationships beyond sequence-level features. **C** XGBoost Feature Importance - XGBoost feature importance (gain metric). Top-ranked ESM embedding dimensions and UMAP features indicate consistent signal patterns across ensemble methods.

Ablation Analysis

Feature ablation experiments compared model performance using all features versus only the top 200 selected features. The results confirmed that although PLMs and dimensionality reduction generate many features, the top 200 features are sufficient for accurate modeling, highlighting the crucial role of feature selection.

ablation analysis summary (all features vs top 200)

	roc_auc (all)	roc_auc (top 200)	delta auc	f1 (all)	f1 (top 200)	delta f1
model						
Logistic Regression	0.914	0.906	-0.007	0.844	0.841	-0.003
Random Forest	0.915	0.918	0.004	0.851	0.848	-0.003
XGBoost	0.915	0.920	0.005	0.850	0.854	0.004

Table 3. Ablation Analysis Summary (All vs Top 200) - Minimal delta AUC (< 0.01) indicates that the top 200 features capture most predictive information.

Hyperparameter Tuning and Test Evaluation

Post-tuning performance showed measurable improvements across all models, with the largest gains observed for Random Forest and XGBoost. Hyperparameter optimization enhanced recall and precision without overfitting.

baseline vs tuned (test set) – with deltas

		roc_auc	f1	accuracy	precision	recall
model	setting					
logistic	baseline	0.9061	0.8410	0.8261	0.7746	0.9199
	tuned	0.9132	0.8475	0.8330	0.7799	0.9279
random forest	baseline	0.9184	0.8482	0.8391	0.8027	0.8993
	tuned	0.9892	0.9465	0.9435	0.8989	0.9995
xgboost	baseline	0.9198	0.8541	0.8399	0.7844	0.9374
	tuned	0.9315	0.8656	0.8524	0.7942	0.9512
delta logistic	tuned-baseline	0.0071	0.0064	0.0069	0.0053	0.0080
delta random forest	tuned-baseline	0.0708	0.0983	0.1044	0.0963	0.1002
delta xgboost	tuned-baseline	0.0117	0.0115	0.0125	0.0098	0.0138

Table 4. Baseline vs Tuned Comparison (Test Set) - *Comparison of baseline and tuned model performance on the test set. Ensemble methods benefit most from tuning, improving AUC by up to +0.07.*

Failure Analysis

A detailed failure analysis was conducted on the tuned XGBoost model, which achieved the highest overall performance. Three representative misclassification cases were examined to better understand model limitations and the nature of difficult-to-predict samples.

1. High-Confidence False Positive (Pred = 0.97, True = 0):
The model predicted binding for a TCR–epitope pair that was experimentally labeled as non-binding. Feature inspection showed that several embedding dimensions and sequence-derived components were highly similar to known binders, causing the model to over-generalize the patterns learned from frequent binding motifs.
2. High-Confidence False Negative (Pred = 0.08, True = 1):
This instance represented an uncommon epitope with limited representation in the training set. The model had difficulty identifying the relevant embedding signals due to sparse examples of this epitope pattern, resulting in low predicted binding probability despite a positive true label.
3. Borderline Misclassification (Pred ≈ 0.52):
This example fell close to the classification threshold, with both binding and non-binding associated features contributing in roughly equal strength. The uncertainty reflects overlapping feature distributions between binding and non-binding cases, which limits clear separation near the decision boundary.

Overall, failure cases highlight the model's sensitivity to data imbalance and rare sequence motifs. Improving the representation of under-sampled epitopes or incorporating structural context (e.g., 3D binding features) could reduce such errors in future iterations.

Discussion

Overall, the supervised learning results confirm that ensemble-based models (RF, XGBoost) outperform linear baselines on the TCR–epitope binding prediction task. The tuned XGBoost model achieved the highest ROC-AUC (0.932) and balanced F1 (0.866), demonstrating superior discriminative ability. Feature importance and ablation analyses revealed that ESM embeddings and PCA/UMAP components are biologically meaningful predictors. Cross-validation results confirmed generalization stability, while tuning further optimized performance across all metrics.

The visual evidence from the ROC and Precision–Recall curves (Figures 1–2) highlights consistent gains from feature selection and tuning. Feature importance plots (Figures 3a–c) reinforce that biologically derived embeddings dominate the decision space. Finally, ablation (Figure 4) and tuning deltas (Figure 5) confirm that reduced, tuned models achieve near-optimal results with less computational complexity.

Ethical Considerations

The primary ethical concerns when working with any biological data are informed consent and risk of de-identification. The providers of the dataset we chose have already accounted for the majority of these concerns by removing the DNA sequences associated with the CDR3 domains, dramatically reducing the ability to identify individuals or even populations. Genetic discrimination has been minimized further by using exclusively the CDR3 binding domain rather than the entire TCR protein sequence.

The 'black-box problem' inherent in unsupervised learning has been addressed in two ways. The first is by providing as much algorithmic transparency as possible into the method used. The second is by using a combination of methods that serve as quality control checks for each other throughout the process in both the supervised and unsupervised sections.

The largest ethical concern of fairness we cannot fully account for is the issue of population underrepresentation. The majority of the available data comes from first world countries with robust research institutions. This has led to an underrepresentation in the antigen epitopes from viruses and cancer markers that primarily impact people in Africa or South America. We have done our best to ensure the model is as unbiased as possible and is readily able to adapt as new information becomes available.

Statement of Work

Thomas Lucas has functioned as the project manager and subject matter expert. His primary responsibilities have been developing the proposal/project, organizing responsibilities/timelines, and compiling all materials related to this report. Liu Liu has served as the unsupervised learning engineer, generating all materials from feature engineering, figure generation, model analysis, and through to final unsupervised learning results. In addition, he has provided the primary quality control checks and review throughout the project. Vishnu Talla performed all activities related to the supervised learning section starting from the features provided from the unsupervised learning section to the final model, including all feature importance, ablation, sensitivity, and failure analysis.

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